

StealthChat: Human-Like Bot for Scam Engagement (Track 2.2)

1. Problem

Scammers detect bots quickly -> conversation ends -> limits research and scam prevention.

2. Approach

- 5 reusable prompt templates (delay, evasive, reverse questioning, casual noise, soft refusal)
- Synthetic dialogue dataset with believability scoring
- Trust-signal matrix modelling human-like behaviour optimisation
- Tactic-signal grid for realism optimisation
- Fully synthetic: no real data used

3. Visual

A simple flow diagram:

Scammer -> Bot -> Detection Shield -> Engagement Loop

Prompt Templates Table:

[6]:					
	name	tactic	description		template
0	Delay Tactic	delay	Simulates realistic delay by stalling before g...	You are chatting with a scammer but you must n...	
1	Evasive Tactic	evasive	Avoids direct answers and pushes the scammer t...	You must sound cooperative but never share con...	
2	Reverse Questioning	reverse_question	Asks questions back so the scammer has to prov...	When the scammer asks for sensitive info, answ...	
3	Casual Human Noise	casual_noise	Adds typos, emojis, and small corrections to a...	Write like a normal person texting on their ph...	
4	Soft Refusal	soft_refusal	Refuses risky actions without fully ending the...	You never share screenshots, codes, or full ID...	

Dialogue Snippets Table:

snippet_id	scenario	role	message	tactic_used	trust_signals_used	believability_score	
0	1	Investment scammer	bot	oh wait my banking app is not loading lol 🤖 gi...	delay + casual_noise	[Typing delay / slow response, Informal langua...	4
1	2	Investment scammer	bot	why do u need the screenshot again? my cousin ...	reverse_question	[Reverse questions, Personal context / backstory]	5
2	3	Romance scammer	bot	sry I didn't see ur msg, was helping my sis wi...	delay + casual_noise	[Typing delay / slow response, Informal langua...	4
3	4	Romance scammer	bot	idk I feel weird sending more pics tbh, my las...	soft_refusal	[Softening / hedging words, Personal context / ...]	5
4	5	Tech support scammer	bot	hmm not sure I see that button... can u explain ...	evasive	[Softening / hedging words]	3
5	6	Tech support scammer	bot	my brother told me never to give remote access...	reverse_question + soft_refusal	[Reverse questions, Personal context / backstory]	4

Trust Matrix:

	Trust Signal	Why it feels human	Example	Implemented via	Risk if overused
0	Typing delay / slow response	Real people do not reply instantly, especially...	Short stalling texts like 'one sec', 'hold on'...	Delay tactic, sending intermediate 'stall' mes...	Too many delays can frustrate scammers and end...
1	Informal language / emojis	Everyday chat apps are full of emojis and slang.	'lol', 'idk tbh 🤔', '👉', '👈', '👉👈'	Casual noise template and light emoji use in r...	Too many emojis feels fake or like a parody.
2	Minor typos and corrections	People make small mistakes and fix them quickly.	'ok wait... *meant', 'sry', 'yea thats fine'	Add 1-2 small typos or corrections per long me...	Heavy errors can reduce clarity and seem scrip...
3	Personal context / backstory	Mentioning family, work, or past experiences f...	'my cousin told me to be careful', 'my last re...	Soft refusal + reverse questioning templates.	Too much story can look like a fixed script.
4	Reverse questions	Real people question things that feel risky.	'Why do u need that again?', 'How do I know th...	Reverse-question template when asked for sensi...	If every reply is a question, it may feel unna...
5	Softening / hedging words	People rarely say 'no' directly; they soften r...	'I don't feel comfy with that', 'maybe later',...	Soft refusal template around risky requests.	Too much hesitation may reduce engagement time.

4. Impact

- Longer scammer engagement
- Safer research interactions
- Better training data
- Zero risk to real victims